Concept-Drift-Aware Credit-Card Fraud Detection Using Online, Imbalance-Aware Classical Machine Learning

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***Abstract*—Payment-card fraud detection is challenged by two coupled difficulties: fraudulent transactions form an extreme minority class, and the patterns that characterise fraud change over time as offenders adapt—a phenomenon known as concept drift. Detectors trained once and evaluated on randomly shuffled data can report strong results while degrading silently in deployment. This synopsis defines the problem of sustaining minority-class fraud detection under realistic, time-ordered conditions using classical (non-deep) machine learning; states the objectives of establishing imbalance-aware baselines, evaluating models with a forward-in-time protocol, and studying drift-triggered adaptation; and identifies a public, time-ordered transaction dataset suitable for the study.**

***Index Terms*—Fraud detection, concept drift, online (incremental) learning, class imbalance, data-stream mining.**

I. PROBLEM STATEMENT

Payment-card fraud accounts for large and rising financial losses worldwide, and financial institutions increasingly depend on machine-learning classifiers to flag fraudulent transactions among billions of legitimate ones [2], [5]. Two properties of this setting make the task persistently difficult. Fraudulent transactions are extremely rare, forming a small fraction of a percent of all activity, so models optimised for overall accuracy tend to under-detect the very cases that matter [3], [12]. In addition, fraudulent behaviour is not static: as offenders adapt to defences and as legitimate spending patterns shift, the statistical relationship between transaction attributes and the fraud label changes over time, a phenomenon known as concept drift [13], [16].

Under concept drift, a classifier that is trained once and then left unchanged gradually loses reliability, catching progressively less fraud, while its reported performance—measured on a fixed, randomly shuffled test set—continues to appear high [14], [16]. A great deal of existing work evaluates fraud detectors on a single static split that mixes past and future transactions, which obscures this temporal decay and gives little insight into how quickly detection quality erodes once distributions shift, or into how a detector should even be assessed when it must operate continuously on an ordered stream of transactions [15], [21]. As a result, it remains unclear to what extent conventional, non-adaptive fraud-detection models actually sustain minority-class detection under realistic, time-ordered conditions, and how the severity of concept drift relates to the loss of detection performance over time.

II. OBJECTIVES

1) To establish classical baseline classifiers (logistic regression, random forest, and gradient boosting) for credit-card fraud detection and to evaluate them with imbalance-aware metrics—PR-AUC, recall at a fixed precision, and false-positive rate—rather than accuracy [3], [4].

2) To implement a prequential (test-then-train) evaluation over a time-ordered transaction stream and to compare a static model, a periodically retrained model, and an online incrementally updated model, quantifying how detection performance degrades over time [15], [22].

3) To integrate classical concept-drift detectors (e.g., ADWIN, DDM) that trigger adaptation upon distribution shift, and to measure the resulting gain in minority-class recall and the model's recovery time after drift [16], [17], [23].

III. DATA SOURCE

The primary dataset is the European credit-card transactions dataset released by the Machine Learning Group, Université Libre de Bruxelles (ULB) / Worldline, publicly available on Kaggle. It contains 284,807 transactions recorded over two days, of which 492 (0.172%) are fraudulent; the 30 features comprise 28 PCA-transformed variables (V1–V28) with the original Time and Amount fields. The academic reference for the dataset is Dal Pozzolo et al. [1]. For a longer temporal horizon, the IEEE-CIS Fraud Detection dataset (590,540 transactions with an explicit timestamp) may serve as a secondary source. Both are open and processable within the project timeframe. Dataset: https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

IV. REFERENCES

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